

SOFTWARE ENGINEERING LAB 3

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Section: G

Scenario No.: 38-Dropshipping inventory & order orchestrator.

Architectural Justification

Architecture Selection

We chose **Microservices Architecture** for the Dropshipping Inventory & Order Orchestrator system.

Architectural Choice

Microservices Architecture: the system is decomposed into independently deployable services — Webhook Ingestion, Order Manager, Supplier Integration, Order & Inventory Data Store, Shipment Tracking, and Notification Service — each communicating over well-defined interfaces rather than sharing a single process.

Reason 1: Independent Scaling

Inbound webhook traffic from the storefront can spike sharply (e.g., flash sales), and NFR-001 requires acknowledging each webhook within 200 ms to avoid a provider retry storm. Supplier inventory-check calls, by contrast, are comparatively slow and rate-limited by third-party APIs. A microservices split lets the Webhook Ingestion service scale horizontally to absorb traffic bursts independently of the Supplier Integration service, so one bottleneck never throttles the other.

Reason 2: Fault Isolation

FR-005 requires retrying failed supplier API calls with backoff before flagging an order for manual review. If a single supplier partner's API is down, that failure must not stall orders destined for other, healthy suppliers, nor delay the webhook acknowledgment path. Isolating Supplier Integration as its own service contains an outage to just the affected supplier's order flow, rather than degrading the whole system as a monolithic design would.

Security Advantage

Each microservice holds only the credentials it needs: the Supplier Integration service manages narrowly-scoped API keys per supplier partner, separate from the Webhook Ingestion service's storefront webhook signing secret. If one service's credentials are compromised, the blast radius is limited to that service's integration rather than exposing every supplier's keys and the storefront's webhook secret from a single shared process, as would be the case in a monolith.

Performance Benefit

The Webhook Ingestion service can immediately persist the incoming order event and return an HTTP 200 acknowledgment within the 200 ms budget, handing the order off asynchronously to the Order Manager and Supplier Integration services for inventory verification and PO dispatch. This decouples webhook acknowledgment latency from downstream processing time, directly satisfying NFR-001 without blocking on slow or rate-limited third-party supplier responses.

COMPONENT DIAGRAM:

